

Task 1: Install MariaDB on your Linux system and verify the installation by running the command to check the MariaDB server version.

Ans :-

Step 1: Install MariaDB Server

```
sudo dnf install -y mariadb-server
```

Step 2: Start and Enable MariaDB

```
sudo systemctl enable --now mariadb
```

Step 3: Check Service Status

```
sudo systemctl status mariadb
```

Expected output:

Active: active (running)

Step 4: Verify MariaDB Version

```
mariadb --version
```

Example output:

```
mariadb Ver 15.1 Distrib 10.5.xx-MariaDB
```

Task 2: Create a new MariaDB database named foodie_app and a table restaurants with columns id (INT, primary key), name (VARCHAR), cuisine (VARCHAR), and rating (FLOAT). Insert 3 sample restaurant records.

Ans :-

Login to MariaDB

```
sudo mariadb
```

Create Database

```
CREATE DATABASE foodie_app;
```

Select Database

```
USE foodie_app;
```

Create Table

```
CREATE TABLE restaurants (
```

```
id INT PRIMARY KEY,  
name VARCHAR(100),  
cuisine VARCHAR(50),  
rating FLOAT  
);
```

Insert Sample Records

```
INSERT INTO restaurants VALUES  
(1,'La Pinoz Pizza','Italian',4.6),  
(2,'Spice Villa','Indian',4.2),  
(3,'Dragon House','Chinese',3.8);
```

Verify Records

```
SELECT * FROM restaurants;
```

Expected Output

```
+----+-----+-----+-----+  
| id | name      | cuisine | rating |  
+----+-----+-----+-----+  
| 1  | La Pinoz Pizza | Italian | 4.6    |  
| 2  | Spice Villa   | Indian  | 4.2    |  
| 3  | Dragon House  | Chinese | 3.8    |  
+----+-----+-----+-----+
```

Task 3: Write and execute a SQL query in MariaDB to find all restaurants in the foodie_app database with a rating greater than 4.0 and display their name and cuisine.

Ans :-

SQL Query

```
SELECT name, cuisine  
FROM restaurants  
WHERE rating > 4.0;
```

Expected Output

name	cuisine
La Pinoz Pizza	Italian
Spice Villa	Indian

Task 4: Grant SELECT and INSERT permissions on the foodie_app database to a new MariaDB user named zomato_user with password zomato123, then test the permissions by connecting as that user and inserting a new restaurant.

Ans :-

Create User

```
CREATE USER 'zomato_user'@'localhost' IDENTIFIED BY 'zomato123';
```

Grant Permissions

```
GRANT SELECT, INSERT
```

```
ON foodie_app.*
```

```
TO 'zomato_user'@'localhost';
```

Apply Changes

```
FLUSH PRIVILEGES;
```

Exit MariaDB

```
EXIT;
```

Login as New User

```
mariadb -u zomato_user -p
```

Password:

```
zomato123
```

Select Database

```
USE foodie_app;
```

Insert New Restaurant

```
INSERT INTO restaurants VALUES
```

(4,'Burger Point','Fast Food',4.3);

Verify Insert

SELECT * FROM restaurants;

Expected Output

id	name	cuisine	rating
1	La Pinoz Pizza	Italian	4.6
2	Spice Villa	Indian	4.2
3	Dragon House	Chinese	3.8
4	Burger Point	Fast Food	4.3

Task 5: Use ChatGPT or Copilot to generate a SQL query that updates the rating of all restaurants with cuisine 'Italian' to 4.5 in your MariaDB table, then run the query and verify the update.
Hint: Paste your prompt and the AI-generated query in your submission along with the result.

Ans :-

Prompt Used

Generate a MariaDB SQL query to update the rating of all restaurants where the cuisine is 'Italian' to 4.5.

AI-Generated Query

UPDATE restaurants

SET rating = 4.5

WHERE cuisine = 'Italian';

Execute Query

UPDATE restaurants

SET rating = 4.5

WHERE cuisine = 'Italian';

Verify Update

```
SELECT * FROM restaurants;
```

Expected Output

id	name	cuisine	rating	
+----+	-----+	-----+	-----+	+
1	La Pinoz Pizza	Italian	4.5	
2	Spice Villa	Indian	4.2	
3	Dragon House	Chinese	3.8	
4	Burger Point	Fast Food	4.3	
+----+	-----+	-----+	-----+	+